Selected W-F: joined forward qualification

16 October 2026 — the original constant-advice domain; no traffic experiment

Status. The user selects W-F and closes J4-W1: merging work has no direct advice dependence at fixed represented physical state. The conditional whole-model forward argument below is delivered at **WF-Q1**, an evidence checkpoint. Physical MP6 and the dependent action, gradient and reference contracts remain open. Earlier proofs keep their stated scope; this is a new deduction for the selected joint member.

1. Primary-domain assumptions and selected expression

Retain the accepted H1 preparation and initially empty finite T1 terminal: fixed $L = L_p + b$, initial footprint $[0, \ell]$, $0 < \ell \le L_p$, initial densities $e_0, a_0 = b_0 \ge 0$, $r = e_0 + a_0$, and $M = \ell(e_0 + 2a_0) > 0$. There are no postdecision external arrivals. One constant common action lies in the original interval $[v_-, v_+]$. In addition to H1's admitted initial and handoff conditions, retain

$$v_+ r < R \le C(v_-), \quad v_+ b_0 \le R_B \le C(v_-), \quad M < k_j b, \quad w, k_j, h > 0. \tag{1}$$

With $K = wk_j$, define

$$C(v) = \frac{Kv}{w+v}, \quad U(f) = \frac{wf}{K-f}, \quad P_j(t) = F_j(t - b/v), \quad q_j = P_j - Y_j, \quad N_j = F_j - Y_j. \tag{2}$$

The far lane retains $Y_B = P_B$. For A/E, $g_j = \dot{Y}_j$ and $\lambda_j = \dot{P}_j$. Write $B(t)$ for shared service, distinct from the far-lane index. Free marks are $u_j = v$; queued marks are $U(g_j)$.

Set fixed $v_\star = v_+$ and $C_\star = C(v_\star)$ once, and use

$$c_A = 1, \quad c_E = c_0 + \chi g_A [u_A - u_E]_+, \quad c_0 = 1 + \delta \ge 1, \quad \chi = \frac{\beta}{C_\star v_\star}, \quad \eta = c_0 - \beta(1 + v_+/w) > 0. \tag{3}$$

$\beta \ge 0$ and all physical coefficients are fixed. χ has units $\mathrm{s}^2/(\text{veh-eq m})$; c_j is dimensionless work per unit mass. The anchor is a fixed coefficient convention, not a variable to recompute with advice or an optimizer's search interval. At fixed g_j, u_j, (3) has zero direct advice derivative. The prior conservative root bound and $c_0 \le c_E \le c_{\max} := c_0 + \beta$ remain sufficient.

2. The retained causal node and recovery account

For a queued E, let $\mathcal{H}(g; a, u)$ denote

$$\mathcal{H}(g; a, u) = g\{c_0 + \chi a[u - U(g)]_+\}, \qquad \partial_g \mathcal{H} \geq \eta > 0. \tag{4}$$

Its unique root for available work x lies between $x/c_{\max}$ and x/c_0, and below $C(v)$. The selected S-E-P branches, on their feasible domains, are:

Current conditions	Actual service
A and E free	$g_A = \lambda_A$, $g_E = \lambda_E$ when $\lambda_A + c_0\lambda_E \leq B$.
A free, E queued	$a = \lambda_A \leq B$, $g_A = a$ and $\mathcal{H}(g_E; a, v) = B - a$.
A and E queued	$g_A = B/2$ and $\mathcal{H}(g_E; B/2, U(B/2)) = B/2$.
A queued, E free	$g_E = \lambda_E$, $g_A = B - c_0\lambda_E$ when $c_0\lambda_E \leq B/2$; otherwise E's offer also has a shortfall.

These are the existing equal-work/unused-service template and free-A priority, not a new allocation. A positive queue follows $\dot{q}_j = \lambda_j - g_j$; at zero, only available mass can pass. An empty stream stays free while that continuation is feasible. A strict loss of free feasibility initiates its queued branch. At a zero hit, test free admission again. Equality alone is not a queue-birth cause. This is the previously selected causal guard, stated as first loss of feasibility for the proof; no queue floor, latch or numerical selection is introduced.

R-E and O-R remain unchanged. With positive acceleration a_r and fixed $m = m_{\text{ref}} = 1$ vehicle-equivalent, use

$$\begin{aligned} c(u) &= \frac{Ku}{w+u}, \quad u_R = U(R), \quad D(\theta; u) = [R - c(\sqrt{(w+u)^2 + 2wa_r\theta} - w)]_+, \\ p(t) &= \frac{1}{m}\int_{[Y_E(t)-m]_+}^{Y_E(t)} D(t - T(\xi); u_E(T(\xi)))d\xi, \quad B(t) = [R - p(t)]_+. \end{aligned} \tag{5}$$

T is the admitted E mass-rank inverse; zero-rate plateaus have zero admission measure. New mass adds profiles and retires oldest weight after the window fills; all retained profiles age physically. Only E deposits new loss. All marked history is predicted local state. Releasing a queue does not reset that history, and retiring history does not remove traffic.

3. The H1 pulse limits E to one waiting episode

Eligible arrivals start at $t_f = (L - \ell)/v$ and end at $t_r = L/v$. Inside this interval H1 implies

$$\lambda_A + \lambda_E = s_v := vr < R, \quad \lambda_A' = -\lambda_E' \in [0, L_A], \quad L_A := v_+^2 e_0/h. \tag{6}$$

Indeed $\lambda_E = ve_{H1}(t - b/v)$ and $\dot{e}_{H1} = -f/h$, $0 \le f \le ve_{H1} \le ve_0$. There are only the front and rear jumps; H1 branch switches preserve continuous rates. No new initial/demand restriction is used in (6).

Initially $B = R$ and A's offer is below R. If

$$W_f(v) := \lambda_A(t_f+) + c_0\lambda_E(t_f+) \le R, \tag{7}$$

all free traffic passes. Since $v \ge u_R$, free E has $D = 0$. Then $B = R$ and $\lambda_A + c_0\lambda_E = s_v + \delta\lambda_E$ is nonincreasing; the free solution persists through the cohort. If (7) fails strictly, A still fits, and E's queued root satisfies $g_E \le (R - \lambda_A)/c_0 < \lambda_E$. E therefore begins waiting at the front with a compatible positive right-hand derivative.

Suppose that positive E queue later first reaches zero, during the pulse. If A is free, its pre-clearance root g_E^Q gives

$$\lambda_E \le g_E^Q \quad \Longrightarrow \quad \lambda_A + c_0\lambda_E \le B. \tag{8}$$

If A remains queued, $\mathcal{H}(g_E^Q; B/2, U(B/2)) = B/2$ gives

$$c_0\lambda_E \le c_0 g_E^Q \le B/2. \tag{9}$$

Thus free E is feasible in either case. New free E profiles are zero. In (5), aging and retirement can only reduce existing loss; hence B is nondecreasing as long as E stays free. Since λ_E decreases, (9) remains valid while A queues. If A clears, its queued service before clearance was $B - c_0\lambda_E$, so free joint admission satisfies (8). Thereafter the joint offered work decreases while B increases. This continuation is self-consistent and cannot activate E again.

Simultaneous A/E clearance also gives joint free feasibility: before the hit $\lambda_A \le B/2$ and $c_0\lambda_E \le B/2$. After the rear there are no arrivals, so neither queue can re-form. Consequently E has at most one waiting episode: it is absent, or begins at the front and eventually clears. This is specific to the retained monotone H1 pulse, neutral initial account and R-E; arbitrary traffic is not covered.

4. Positive budget and finite A switching

From (3), admitted E rate is at most $G = R/c_0$. The unchanged profile is nonincreasing in age and mark, bounded by R, and strictly less than R at positive age when $a_r > 0$. The earlier rearrangement argument gives

$$p \le p_{\max} := \frac{G}{m}\int_0^{m/G} D(\theta;0)d\theta < R, \qquad B \ge B_{\min} := R - p_{\max} > 0. \tag{10}$$

This is derived, not an added floor. $B_{\min}$ is common to all actions for fixed physical parameters. Its value can be very small.

The injection, aging and retirement update in the earlier O-R note yields an a.e. Lipschitz bound

$$|\dot{B}| \le L_B := 2GR/m + a_r k_j. \tag{11}$$

The retained mass is at most m, $|D_\theta| \le a_r k_j$, and each boundary term is at most GR/m. No inverse-rate lower bound is needed. This bound permits speed-mark jumps at queue events.

Before E clears, any A birth inside the pulse occurs when the free offer first loses feasibility, at $\lambda_A = B$. Switching to queued cooperation gives $\dot{q}_A = B/2 \ge B_{\min}/2 > 0$. If A subsequently clears while E still queues, continuity of the pre-clearance derivative gives $\lambda_A \le B/2$. For $y = B - 2\lambda_A$, its value at birth is $-B \le -B_{\min}$ and its value at clearance is nonnegative. By (6),(11), a birth-to-clearance interval is at least

$$\frac{B_{\min}}{L_B + 2L_A}. \tag{12}$$

At A clearance, the free margin $x = B - \lambda_A$ is at least $B_{\min}/2$. The next A birth needs $x = 0$; thus a clearance-to-birth interval is at least $B_{\min}/[2(L_B + L_A)]$. A common smaller bound is

$$\tau_0 = \frac{B_{\min}}{2(L_B + 2L_A)} > 0. \tag{13}$$

The bound applies to consecutive A events while E queues and the arrival pulse continues. E clearance and the one external pulse end add finitely many exceptions. Once E is free, the previous section excludes further E births, and a cleared A cannot re-form because joint free work fits and decreases. After the rear, positive queues only drain.

Therefore A events cannot accumulate in finite time. Tangential zero hits use the already selected free-admission test; the post-clearance margin still holds. The positive separation is a proof consequence, not an implemented dwell timer or additional historical allocation preference.

5. Global continuation, storage and complete clearance

Causal construction. On each valid branch, (4) and the bounded derivative of U on $[0, R]$ make flows and speed marks locally Lipschitz in B and continuous arrival data. The earlier moving-cutoff O-R estimate for histories with the same past is

$$|\Delta p(t)| \le \frac{1}{m}\left(5R\int |\Delta g_E|ds + k_j G\int |\Delta u_E|ds\right). \tag{14}$$

This gives short-interval Volterra contraction on a valid branch, including cutoff movement and old mark jumps. Continue to its next guard. At the front E shortfall, queue birth has the strict compatible sign shown above. At A birth, the new right-hand derivative is positive. A zero hit admits free A; E zero hits admit free E by (8)–(9). Simultaneous clearance admits both. Equality contacts do not themselves cause birth; keep the maximal feasible free continuation, as the selected guard requires. Point values of flux at a switch do not alter continuous cumulative mass.

Boundedness prevents finite-time blow-up within branches. Before the rear, E switches at most twice and repeated A events are separated by (13), so branch continuation cannot require infinitely many switches on that finite pulse. After the rear, no new queues can be born; there are at most two clearances. This proves a unique causal, piecewise-branch continuation for every original advice under (1)–(3) and the selected causal convention. No transversality assumption or advice subinterval is needed for forward existence; smooth dependence on advice is a separate issue.

If $Q = q_A + q_E > 0$ after t_r, the node uses the available work and $g_A + g_E \ge B/c_{\max} \ge B_{\min}/c_{\max}$. Therefore

$$T_{\text{clear}}(v) \le \frac{L}{v} + \frac{c_{\max}Q(L/v)}{B_{\min}} \le T_{\max} := \frac{L}{v_-} + \frac{c_{\max}M}{B_{\min}} < \infty. \tag{15}$$

Thus the entire cohort clears. Afterwards profiles simply age out within the unchanged finite support. The bound does not assert practical runtime.

H1 entrance flows fit $C(v)$ and $M < k_j b$. The retained T1 vacancy proof therefore protects the handoff for these admissible departures. Physical stock remains $N_j = F_j - Y_j$, with $0 \le Y_j \le P_j \le F_j$. If s_{occ} is upstream occupied length and $\rho_0 = e_0 + 2a_0$,

$$s_{\text{occ}}\rho_0 + \sum_j N_j + \sum_j Y_j = M, \qquad J(v) = \frac{M(L-\ell/2)}{v} + \int_0^\infty (q_A + q_E)dt. \tag{16}$$

No work weight changes traffic mass, and post-exit recovery is not local vehicle-time. The result is not a general invariant LTM network theorem.

6. Information, negative controls and sensitivity limits

Known M, ℓ, E_0 and continuing symmetry reconstruct

$$e_0 = E_0/\ell, \qquad a_0 = b_0 = (M - E_0)/(2\ell). \tag{17}$$

Together with known empty terminal/account and fixed parameters, uniqueness now gives identical full forward prediction for A+E and explicit lane G for every original advice. Their predicted history is not an observation. This does not imply that the current meter alone is sufficient dynamic state. The earlier pre-decision collision stays upstream of an empty terminal and is unchanged; no old action/gradient certificate is transferred.

A sharper negative control. If $\delta = 0$, (7) always holds since $\lambda_A + \lambda_E = s_v < R$. Hence no terminal queues or recovery are activated, regardless of β: free A and E have equal marks. The entire primary objective is $M(L - \ell/2)/v$, so fastest advice is common to all preparations. No E is another such control. Immediate maneuver work is therefore the trigger in this particular underloaded model, not a universally necessary congestion mechanism.

Front threshold. With fixed $z_f = L_p - \ell$, H1 comparison gives $e_f(v) = e_{H1}(z_f; v)$ nondecreasing in v: $k_c(v)$ decreases, so the receiving-limited reduction becomes no stronger. Thus

$$W_f(v) = v\{r + \delta e_f(v)\} \text{ is strictly increasing for } r > 0. \tag{18}$$

There is at most one transition between a free and initially queued cohort as advice increases. It may lie outside the action interval. Equality belongs to free passage under the selected guard.

Forward uniqueness does not imply differentiability at that threshold. If it is interior, with $a = \lambda_A > 0$, $e = \lambda_E > 0$ and $a + c_0 e = R$, then for $\chi > 0$,

$$\mathcal{H}(e; a, v) - (R - a) = \chi a e\{v - U(e)\} > 0. \tag{19}$$

The queued-root limit is strictly less than the free flow e. Speed marks and later recovery can therefore change abruptly across activation. Equation (19) is a local flux statement, not a proof of a jump in complete J. An ordinary smooth derivative across it must not be presumed. Stable event sensitivities, grazing/boundary cases, cutoff motion and full-objective tests remain in the next evidence work; no AD/FD run is performed here.

Scope/checkpoint. The planned forward block is complete at WF-Q1. The older regular A-delay/active-renewal witness is exactly preserved at the fixed reference advice. Physical speed identification, work/recovery partition, joint scales/applied MP6 and independent usefulness remain open. Carey's work interpretation, Leclercq's isolated recovery context and Jin's separate discontinuous-boundary model motivate ingredients, not this proof. Step 3 is open; Step 4 and implementation/experiments remain unopened.